

Final project

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Introduction

This report follows the required steps for a longitudinal social network analysis using stochastic actor-oriented models (SAOMs) implemented in RSiena. We analyze directed “best friends” nominations in a Dutch secondary school classroom across three waves (**V**, **X**, **Y**) from Knecht’s adolescent networks study (Knecht, 2004; <https://doi.org/10.17026/DANS-Z9B-H2BP>).

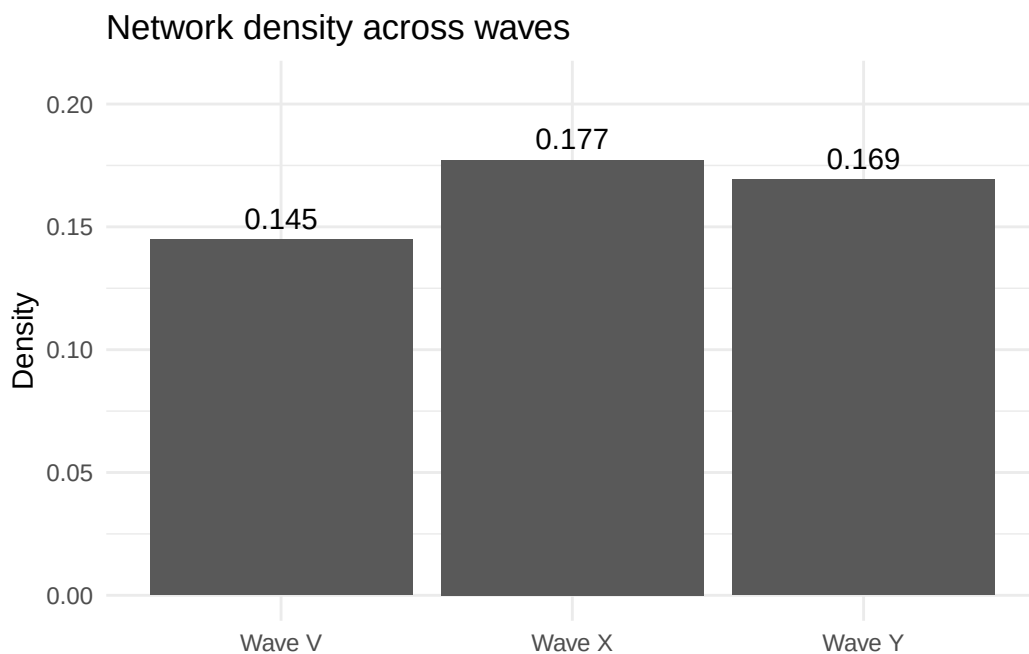
We examine whether (i) sex homophily shapes best-friend nominations (dyadic hypothesis), (ii) triadic closure/transitivity drives clustering (triadic hypothesis), and (iii) a node-level popularity mechanism affects nominations (indegree-related popularity; **inPop**). We estimate an initial specification (Model 1) and then update it once based on goodness-of-fit diagnostics (Model 2, final). SAOM estimation is carried out on a constant node set (pupils observed in all three waves).

1. Network Descriptives

We focus on classroom **09f** (variable `schoolnr`). This classroom was selected because it provides at least three observed waves and shows sufficient stability (composition and tie Jaccard) to support SAOM estimation while still exhibiting meaningful change over time. Although the original dataset contains four survey waves (W, V, X, Y), we restrict the analysis to waves V, X, and Y. Including wave W would reduce the set of pupils observed in all waves (a constant node set is required for SAOM estimation), which would substantially shrink the network used in the longitudinal model. Therefore, we focus on the three waves that preserve sufficient overlap and comparability for SAOM estimation.

Figure 1 summarizes network density across waves (using the observed classroom composition in each wave).

Figure 1: Network density across waves



These values are proportions of possible friendships that exist. So for Wave V, only about 14.5% of all possible ties were actually reported. Wave X is slightly denser (about 17.7%), and Wave Y is very similar (about 16.9%). Networks with density < 0.2 are considered sparse, which is very common in social networks because not everyone is friends with everyone else. Table 1 reports basic descriptives of the best-friends network across waves.

Table 1: Basic descriptives of the best-friends network across waves (observed classroom composition per wave)

Wave	Nodes	Density	DensityChangePct
Wave V	30	0.145	NA
Wave X	28	0.177	22.386
Wave Y	26	0.169	-4.524

Across the three waves, classroom composition changes slightly (30 pupils in V, 28 in X, 26 in Y). Network density is low (0.145–0.177), which is typical for “best friends” nomination networks.

SAOM estimation requires a constant set of actors. Therefore, after describing the observed networks above, we (i) quantify composition stability using the node-set Jaccard index, (ii) quantify tie turnover using the tie Jaccard index computed on a constant node set, and (iii) restrict SAOM estimation to pupils observed in all three waves (the intersection sample).

Missing or empty nomination slots: Each respondent could nominate up to 12 classmates. Nomination entries coded as 98/99 are treated as no nomination, empty slot and are not converted into ties. Nominations to IDs not present in the classroom roster of that wave are discarded. All remaining non-nominated dyads are coded as 0 ties.

Table 2: Jaccard stability of the node set (classroom composition)

Wave_Comparison	Jaccard_NodeSet
Wave V $\rightarrow$ Wave X	0.933
Wave X $\rightarrow$ Wave Y	0.862

Table 3: Number of students present in all three waves (intersection)

Students_in_all_three_waves
25

Table 4: Basic descriptives of the best-friends network on the constant node set (intersection sample).

Wave	Nodes	Density
Wave V	25	0.142
Wave X	25	0.192
Wave Y	25	0.173

The intersection sample contains **25** pupils and is the sample used for SAOM estimation..

Table 5: Jaccard stability of ties on the constant node set (intersection sample)

Wave_Comparison	Jaccard_Ties
Wave V $\rightarrow$ Wave X	0.242
Wave X $\rightarrow$ Wave Y	0.470

The table above reports tie Jaccard stability (overlapping friendships) between consecutive waves on the constant node set used for SAOM. Higher values indicate more stable friendship relations across time, supporting the use of longitudinal modelling. The Jaccard indices above quantify how stable the classroom composition is across waves. Because SAOMs require a constant node set, the subsequent SAOM analysis will be run on the students who are present in all three waves (the intersection).

2. Dyadic Hypothesis

For the analysis of dyadic characteristics in the “Best friends” network, we formulated the following hypothesis: students are more likely to nominate as best friends those peers who are of the same sex. This hypothesis is based on the well-established principle of sex homophily in adolescent social networks, where boys tend to form friendships with other boys, and girls with other girls. Theoretical considerations, such as social identity theory, suggest that individuals prefer closer ties with peers who share salient characteristics like sex, which can also facilitate participation in similar classroom activities or interests. Empirical studies in classroom social networks consistently report high levels of sex-based homophily (e.g., Snijders et al., 2010). To test sex homophily in RSiena, we use the **sameX** effect with the actor covariate **sex_cov**. A positive **sameX** parameter indicates that pupils are more likely to nominate peers of the same sex as best friends.

As a descriptive complement without presenting large similarity matrices, we report the share of observed nominations that go to same-sex peers in each wave computed on the constant node set used in the SAOM.

Table 6: Share of same-sex best-friend nominations by wave (constant node set).

Wave	Share_same_sex
Wave V	0.800
Wave X	0.730
Wave Y	0.769

3. Triadic Hypothesis

For the analysis of triadic characteristics in the “Best friends” network, we formulated the following hypothesis: friendship ties are likely to exhibit transitivity, meaning that if student A nominates student B as a best friend, and student B nominates student C, then student A is also likely to nominate student C. This hypothesis is grounded in the principle of triadic closure, a common phenomenon in social networks, where “friends of friends tend to become friends” (Granovetter, 1973; Snijders et al., 2010). Theoretical explanations suggest that transitive ties increase trust, reduce social tension, and facilitate cooperation within a group. In classroom networks, transitivity is often observed because students share activities, classes, and social environments, making it more likely that friendship clusters form. To operationalize this triadic hypothesis in RSiena, we use the transTrip effect, which captures triadic closure directly within the SAOM. A positive transTrip parameter indicates that the presence of A->B and B->C increases the likelihood of A->C forming over time, reflecting clustering in the classroom network. As a descriptive check, we also compute and report global transitivity (triadic closure) for each wave on the constant node set used for the SAOM.

Table 7 reports global transitivity (triadic closure) for each wave.

Table 7: Global transitivity by wave

Wave	Transitivity
Wave V	0.444
Wave X	0.590
Wave Y	0.484

The transitivity of the “Best friends” network, which measures the tendency of friends-of-friends to also be friends (triadic closure), shows a moderate to high

level of clustering across the three waves. In Wave V, the transitivity is 0.444, indicating that a substantial share of possible triads are closed. Wave X exhibits a transitivity of 0.59, reflecting the tendency for friendship triads to form. In Wave Y, the transitivity is 0.484, suggesting that the level of triadic closure remains in a similar range over time. These results support the hypothesis that friendship formation in the classroom is characterized by transitive relationships, consistent with established findings in adolescent social networks where social groups tend to form tightly connected clusters

4. Node Attribute Hypothesis

For the analysis of node-level characteristics in the “Best friends” network, we formulated the following hypothesis: students with higher levels of popularity (indegree) are more likely to be nominated as best friends by their peers. This hypothesis is based on the idea that individuals with high social status or visibility in the classroom attract more friendship nominations, consistent with the principle of preferential attachment in social networks. Empirical studies in adolescent classrooms indicate that popular students tend to receive more friendship ties and maintain them over time (e.g., Snijders et al., 2010). To operationalize this hypothesis, a popularity score was assigned to each student based on the total number of received best-friend nominations (indegree). This score can then be used to capture popularity dynamics in a SAOM (here via the inPop effect) to test whether higher popularity increases the likelihood of receiving additional nominations over time.

Figure 2: Mean popularity (indegree) across waves, constant node set used for SAOM ($n = 25$).

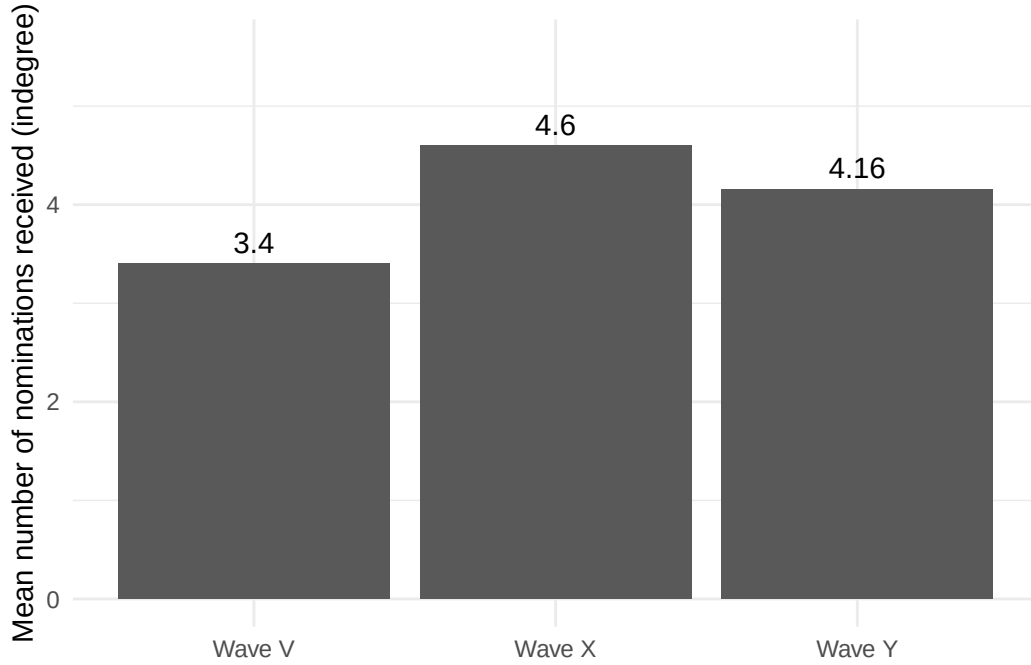


Table 8: Popularity descriptives (indegree) by wave, constant node set used for SAOM.

Wave	Mean	Min	Max
Wave V	3.40	0	6
Wave X	4.60	0	7
Wave Y	4.16	1	7

For the node-level analysis, popularity was calculated as indegree, the total number of best-friend nominations each pupil received in each wave. Because SAOM estimation requires a constant node set, all popularity descriptives reported here are computed on the same intersection sample used in the model ($n = 25$). On this constant node set, mean popularity is 3.40 in Wave V, 4.60 in Wave X, and 4.16 in Wave Y (Figure 2), suggesting a temporary increase in received nominations in Wave X followed by a slight decrease in Wave Y. The range of popularity also indicates heterogeneity in nomination concentration within the classroom, with minima and maxima of 0 to 6 in Wave V, 0 to 7 in Wave X, and 1 to 7 in Wave Y.

5. Control Variables

In our analysis of the “Best friends” network, several control variables were included to account for factors that might influence friendship formation and maintenance:

Ego-level variables: These describe characteristics of the student sending the friendship nomination. For example, we include outdegree activity (outAct) to account for heterogeneity in pupils’ tendency to send nominations (i.e., some students nominate more peers than others).

Alter-level variables: These describe characteristics of the student receiving the nomination. For instance, we include indegree-related popularity (inPop) to capture whether pupils who already receive many nominations become more/less likely to receive additional nominations over time (an endogenous degree-related mechanism).

Dyadic variables: These variables describe pairwise relationships between students. For example, sex homophily is modeled via sameX using the actor covariate sex_cov, capturing the tendency for students to nominate same-sex peers as best friends.

Structural/network variables: These account for endogenous network processes, such as outdegree (density), which captures the baseline sparsity of the network, reciprocity, which captures the tendency for a nomination to be returned by the alter, and transitivity (transTrip), which measures the tendency for friends of friends to become friends.

Together, these control variables allow the stochastic actor-oriented model to isolate the effects of individual, dyadic, and structural factors on the evolution of the friendship network over time.

Table 9 lists the control variables used in the analysis..

Table 9: Control variables to be included in the analysis

Variable	Type	Description
Outdegree (density)	Structural (baseline)	Baseline tendency toward a sparse nomination network.
Reciprocity	Dyadic/structural (endogenous)	Tendency to reciprocate a best-friend nomination.
Sex homophily (sameX)	Dyadic (based on actor covariate)	Preference for same-sex nominations (computed from sex_cov via sameX).

Variable	Type	Description
Transitivity (transTrip)	Triadic/structural (endogenous)	Friends-of-friends closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$.
Indegree-related popularity (inPop)	Degree-related / endogenous	Whether higher-indegree pupils become more/less likely to receive additional nominations over time (endogenous popularity).
Outdegree activity (outAct)	Degree-related / endogenous	Heterogeneity in pupils' tendency to send nominations beyond baseline effects.
Reciprocity activity (reciAct)	Degree-related / endogenous	Heterogeneity related to reciprocating nominations; added to improve outdegree fit (Model 2).

6. Longitudinal Plots

We visualize the best-friends network over time (waves V, X, Y) using fixed node positions and node colors by popularity (indegree; nominations received), recalculated in each wave on the constant node set used for SAOM ($n = 25$). Fixed positions allow direct visual comparison of tie changes and shifts in popularity without layout-induced movement. Figure 3, Figure 4, and Figure 5 suggest two descriptive patterns that motivate our SAOM specification. First, nominations concentrate around a small subset of pupils (lighter node colors), indicating heterogeneity in receiving ties (popularity) across waves. Second, the network displays clustered local structure: many ties appear embedded in small groups, and several ties are part of “friend-of-a-friend” patterns, consistent with a potential triadic closure mechanism. At the same time, there is noticeable rewiring across waves especially between V and X suggesting meaningful turnover in nominations while maintaining a broadly similar clustered structure.

Figure 3: Best-friends network in Wave V (fixed node positions; node color indicates popularity/indegree).

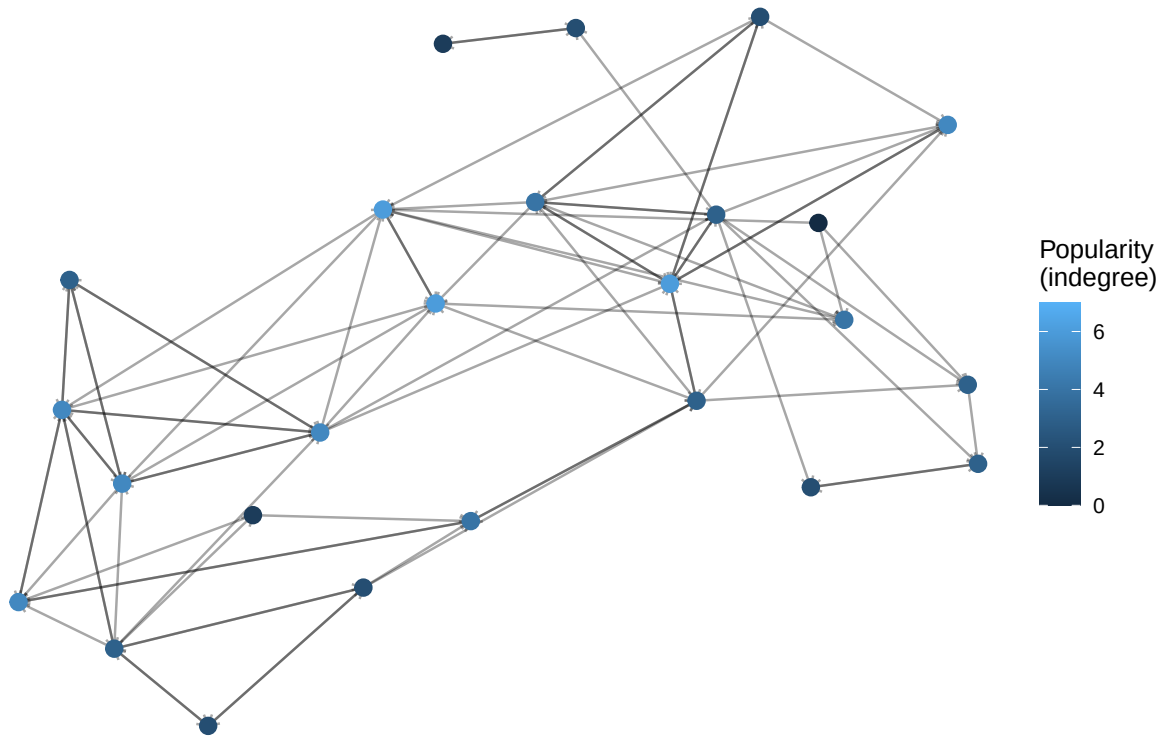


Figure 4: Best-friends network in Wave X (fixed node positions; node color indicates popularity/indegree).

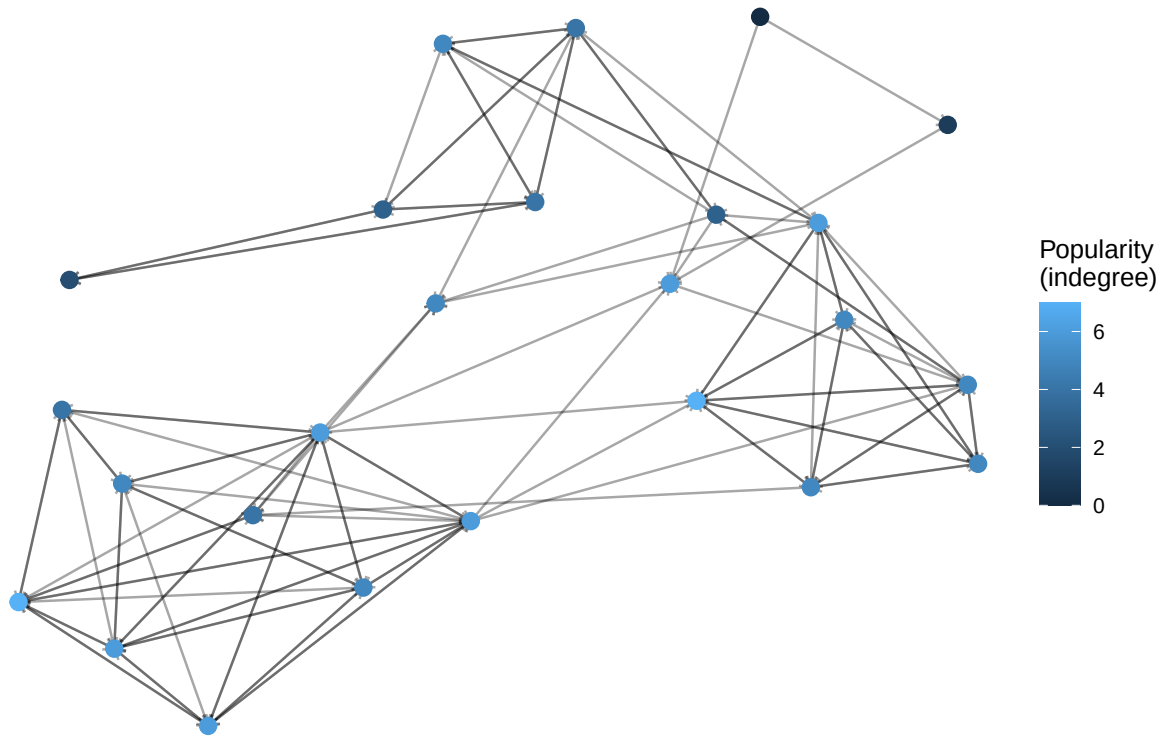
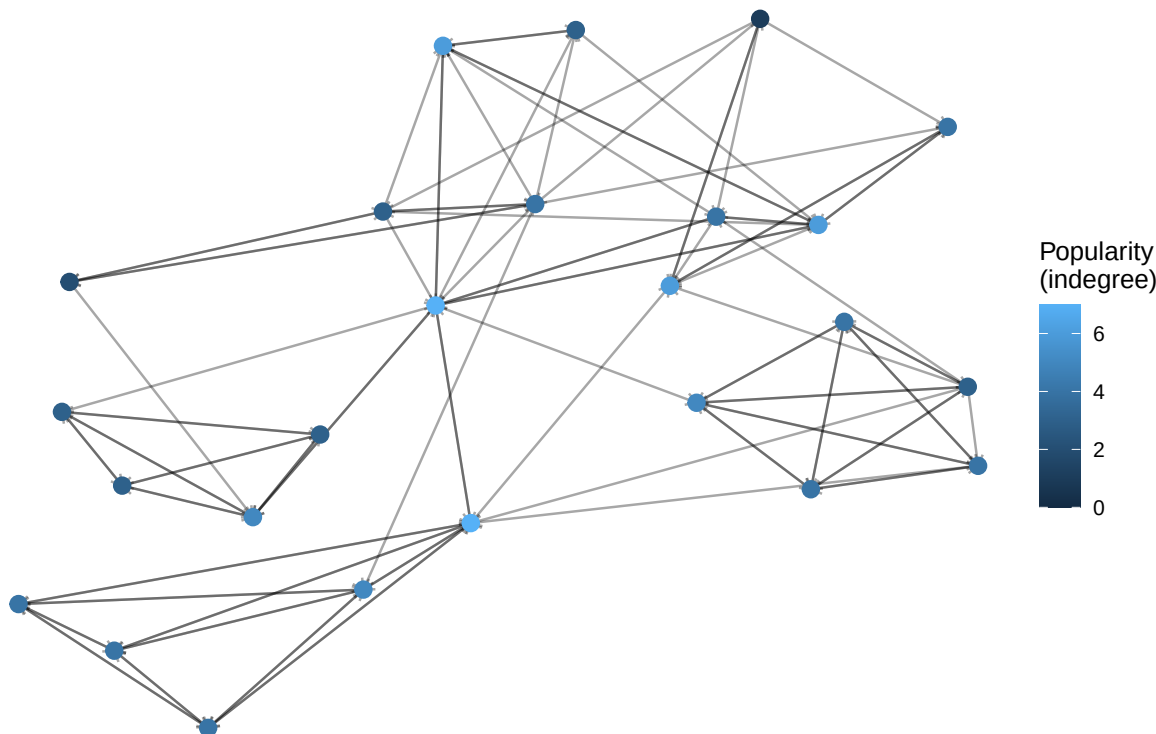


Figure 5: Best-friends network in Wave Y (fixed node positions; node color indicates popularity/indegree).



7. Model Specification

In this project, we test three hypotheses using a stochastic actor-oriented model (SAOM) in RSiena. We operationalize each hypothesis by selecting the corresponding RSiena effect(s). Table 10 shows how each hypothesis is translated into a SAOM (RSiena) effect.

Table 10: Operationalization of hypotheses as RSiena effects.

Hypothesis	RSiena_effect	Expc_di- rec	Interpretation
H1 (Dyadic): sex homophily	sameX (interaction1 = sex_cov)	Posi- tive	Ties are more likely between pupils of the same sex.
H2 (Triadic): transitivity	transTrip	Posi- tive	Friend-of-a-friend closure: $A \rightarrow B$ and $B \rightarrow C$ increases $A \rightarrow C$.

Hypothesis	RSiena_effect	Expc_di-	Interpretation
		rec	
H3 (Node-level): popularity reinforcement	inPop (indegree popularity)	Positive	Pupils who already receive many nominations are more likely to receive additional nominations over time.

Sex homophily is modeled with the `sameX` effect using the sex covariate (`interaction1 = "sex_cov"`). A positive parameter indicates that pupils are more likely to nominate same-sex peers as best friends. Transitivity is captured by `transTrip`, where a positive parameter indicates triadic closure (“friends of friends become friends”). Our node-level hypothesis about popularity is operationalized with `inPop` (indegree popularity), an endogenous effect that tests whether pupils who already receive many nominations tend to attract additional nominations over time. Finally, we include `outAct` (outdegree activity) as an endogenous structural control to account for heterogeneity in pupils’ tendency to send nominations beyond the focal hypotheses.

8. Model Estimation

We estimate SAOMs in RSiena on the constant node set (intersection sample; 25) using the aligned adjacency matrices (`netV_saom`, `netX_saom`, `netY_saom`). RSiena’s default network objective function already includes baseline structural terms (outdegree/density and reciprocity).

Model 1 adds our hypothesis-driven effects (`sameX(sex_cov)`, `transTrip`, `inPop`) and a standard degree-heterogeneity control (`outAct`). If convergence is not reached, we increase estimation effort (`n3`) and continue from previous estimates (`prevAns`). Based on GoF diagnostics, we updated the specification by adding `reciAct`.

Estimation used a fixed random seed (`seed = 12345`) to support reproducibility. Convergence was assessed using RSiena’s t-ratios for convergence, with `tconv.max` below 0.25 indicating acceptable convergence; the final run achieved `tconv.max = 0.193` (Table 11). Rate parameters (Table 12) indicate substantially more opportunities for tie change between $V \rightarrow X$ than $X \rightarrow Y$, consistent with greater turnover earlier in the observation window

Table 11: Convergence diagnostics for the final estimation run.

Metric	Value
<code>tconv.max</code>	0.193
<code>max(tconv)</code>	0.041

Metric	Value
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Table 12: Rate parameters (opportunities for change) between waves.

Period	Rate	SE
$V \rightarrow X$	24.485	10.152
$X \rightarrow Y$	11.226	4.130

Rate parameters summarize the average amount of change (opportunities for tie updates) between waves; higher rates indicate more frequent network updating.

Table 13 shows the parameter estimates for the final model (Model 2).

Table 13: RSiena parameter estimates for the final model.

Effect	Estimate	SE	t-ratio
outdegree (density)	-0.814	0.430	-1.895
reciprocity	3.604	0.586	6.150
transitive triplets	0.466	0.052	8.988
indegree - popularity	-0.336	0.079	-4.274
outdegree - activity	0.127	0.043	2.929
$\text{rec.degree}^{\wedge}(1/1)$ - activity	-0.460	0.109	-4.218
same sex_cov	0.246	0.124	1.981

9. Model Fit

Goodness-of-fit (GoF) assesses whether the fitted SAOM can reproduce important structural features of the observed network. We simulate multiple networks from the fitted model and compare observed network statistics to the distribution of the same statistics in the simulated networks. If the observed statistics are not extreme relative to the simulated distribution (and the Monte Carlo p-values are not close to 0 or 1), the model is considered to fit those features reasonably well.

Here, we evaluate GoF for the indegree and outdegree distributions and for the triad census. In Model 1, the outdegree distribution shows clear misfit (Figure 7; $p = 0.000$), motivating a specification update. After adding reciAct (Model 2, final), fit improves substantially for both degree distributions: indegree GoF becomes acceptable (Figure 8; $p = 0.408$) and outdegree GoF improves (Figure 9; $p = 0.172$), indicating that the updated model better reproduces observed nomination activity patterns and their heterogeneity.

However, the triad census GoF remains poor in the final model (Figure 10; $p = 0.012$), meaning that even though the model includes transTrip, it does not fully reproduce the observed distribution of triad types. This residual misfit suggests that conclusions about triadic mechanisms should be stated as evidence for a general closure tendency within the model, rather than as a complete representation of all higher-order clustering patterns in the classroom

Figure 6: Goodness-of-fit for the indegree distribution (Model 1).

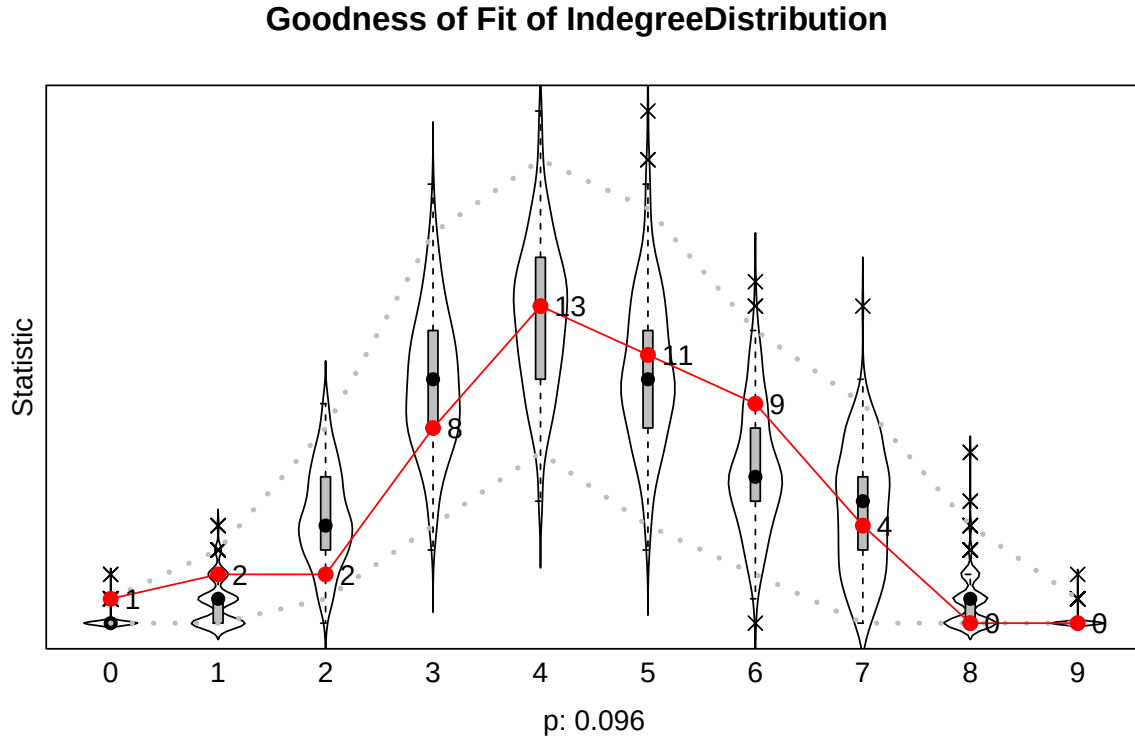


Figure 7: Goodness-of-fit for the outdegree distribution (Model 1).

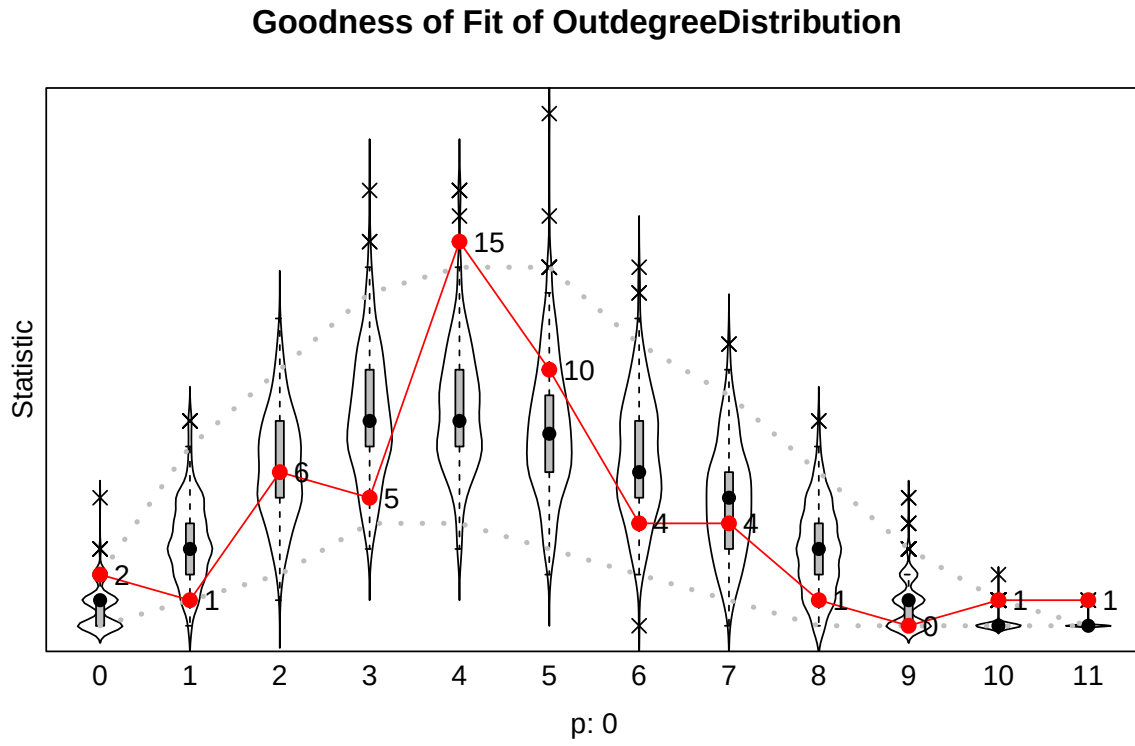


Figure 8: Goodness-of-fit for the indegree distribution (Model 2, final).

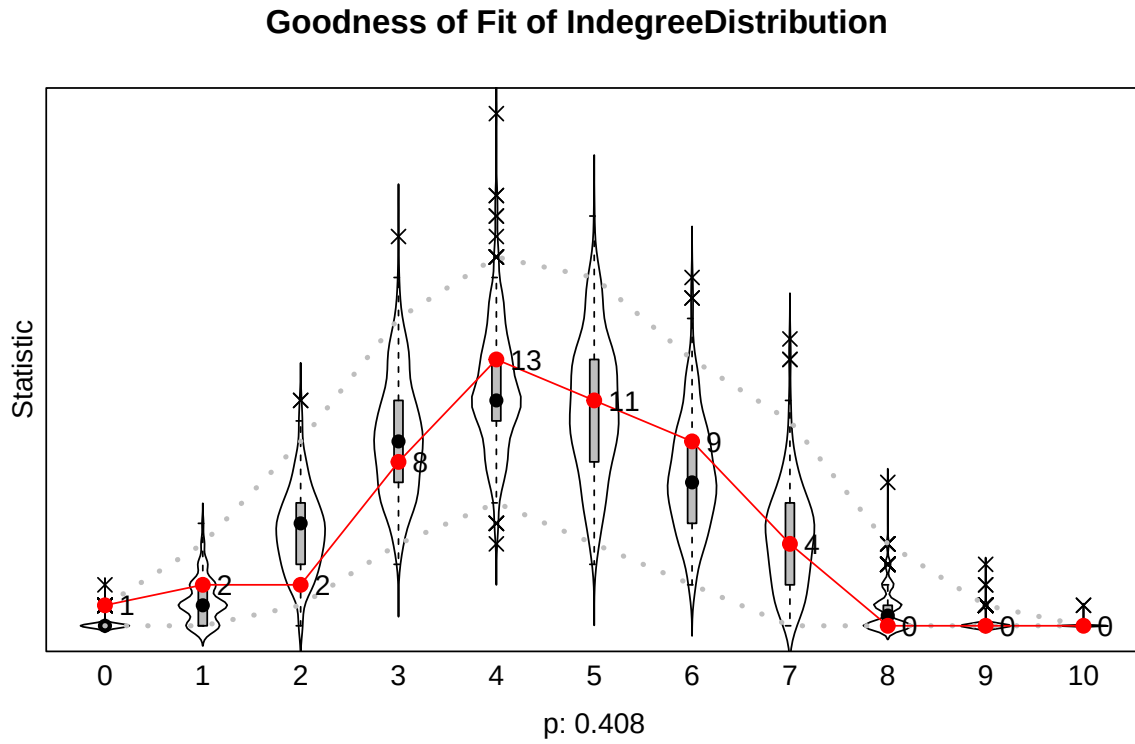


Figure 9: Goodness-of-fit for the outdegree distribution (Model 2, final).

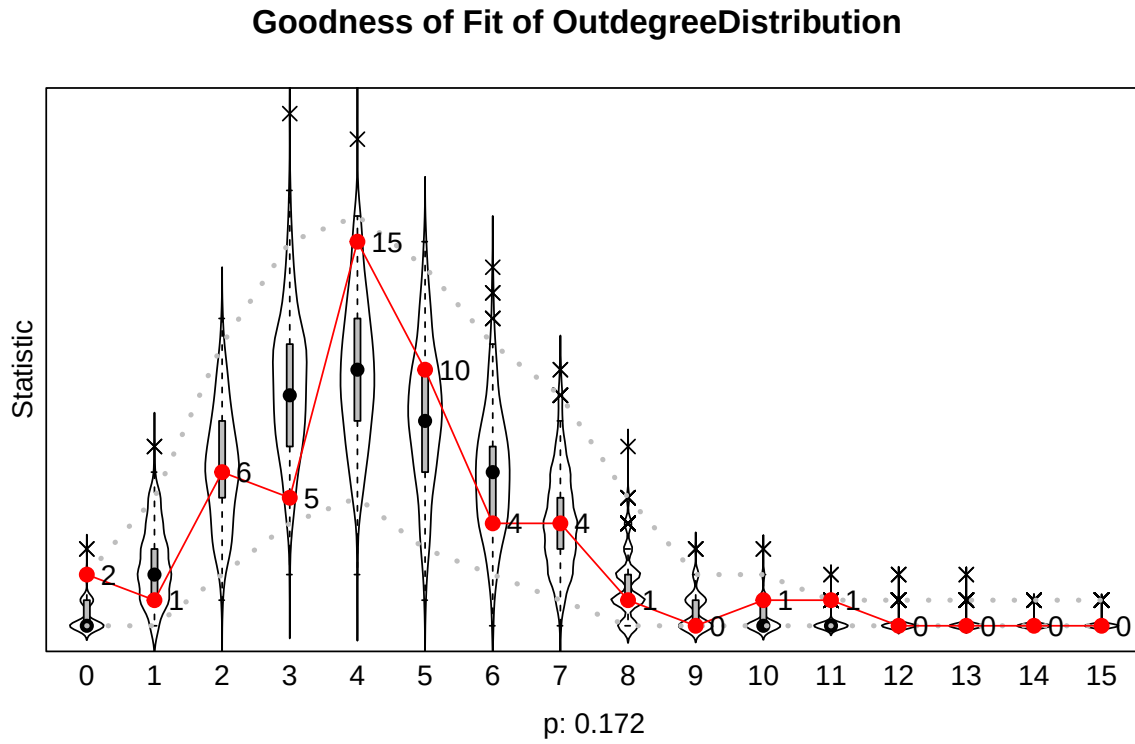


Figure 10: Goodness-of-fit for the triad census (Model 2, final).

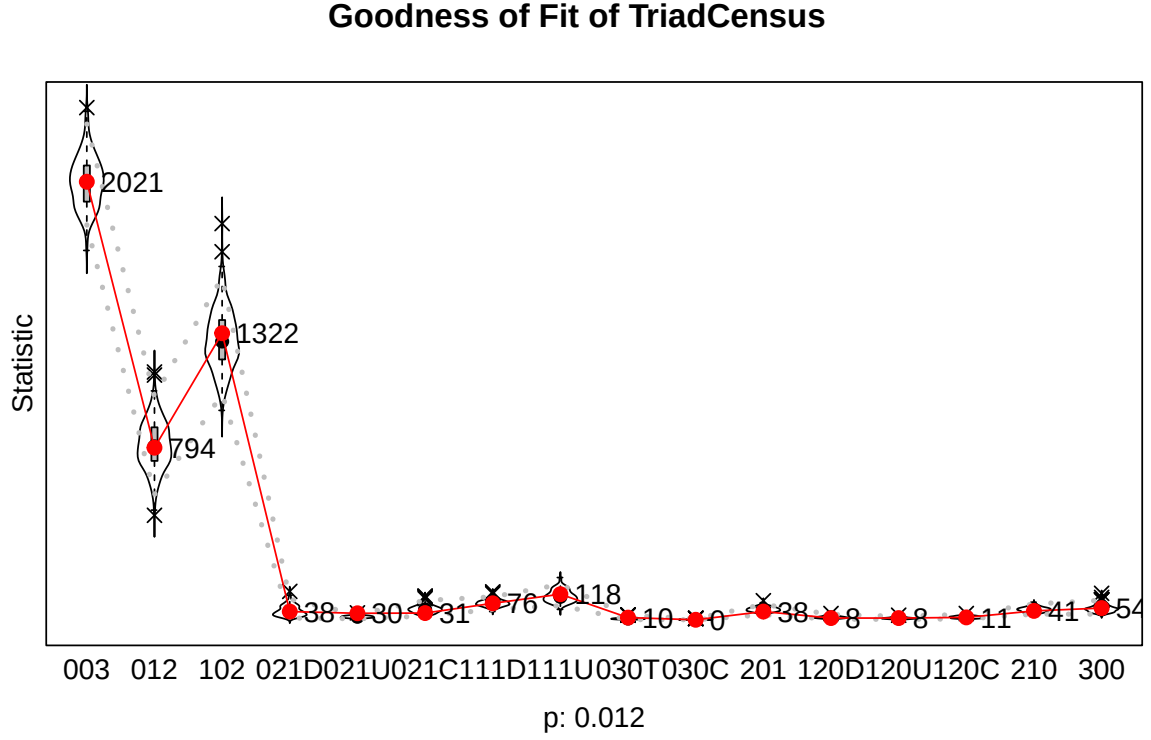


Table 14: Comparison of GoF Monte Carlo p-values (Model 1 vs Model 2).

Statistic	P_Model1	P_Model2
Indegree distribution	0.096	0.408
Outdegree distribution	0.000	0.172
Triad census	0.000	0.012

Table 15: Key estimates comparison, Model 1 vs Model 2.

Effect	Esti- mate_M1	SE_M1	t_M1	Esti- mate_M2	SE_M2	t_M2
indegree - popularity	-0.376	0.079	-4.760	-0.336	0.079	-4.274
outdegree - activity	-0.044	0.023	-1.908	0.127	0.043	2.929
reciprocity	1.641	0.205	7.992	3.604	0.586	6.150
same sex_cov	0.266	0.112	2.388	0.246	0.124	1.981
transitive triplets	0.350	0.038	9.158	0.466	0.052	8.988

10. Discussion and Limits

We modeled best-friend nominations in classroom 09f across three waves (V, X, Y) using a stochastic actor-oriented model (SAOM) estimated on the constant node set (intersection sample; $n = 25$). We tested (H1) sex homophily (sameX), (H2) transitivity (transTrip), and (H3) a popularity-related mechanism (inPop). The final specification (Model 2) converged successfully, and the estimates can therefore be interpreted as evidence about tie formation and maintenance mechanisms conditional on the included controls.

H1 (sex homophily): The sameX estimate is positive but borderline (Estimate = 0.246, $t = 1.981$; Table 13). This shows weak evidence that actors prefer same-sex nominations, conditional on strong endogenous mechanisms such as reciprocity and transitive closure. Given the small intersection sample ($n = 25$) and the fact that many nominations are already structured by reciprocity and local clustering, sex homophily may be partially absorbed by these competing network processes, so we treat H1 as suggestive rather than robust.

H2 (transitivity): Transitive triplets are strongly positive (Estimate = 0.466, SE = 0.052, $t = 8.988$; Table 13), indicating that, at micro-steps, pupils are more likely to create or maintain ties that close transitive structures ($A \rightarrow B$ and $B \rightarrow C$ makes $A \rightarrow C$ more attractive), conditional on reciprocity and degree-related effects. This is consistent with the clustered structure visible in the longitudinal network plots (Figures 3–5). At the same time, the triad census GoF remains poor (Figure 10; $p = 0.012$), so we interpret the result as evidence for a general closure tendency rather than claiming that the model fully captures all observed triadic configurations.

H3 (popularity reinforcement): Contrary to the hypothesis, indegree-related popularity is significantly negative (Estimate = -0.336 , SE = 0.079, $t = -4.274$; Table 13). In SAOM terms, conditional on other mechanisms, ties to already highly nominated pupils become less attractive at micro-steps. One plausible interpretation in this setting is that “best-friend” choices may rotate rather than accumulate (e.g., pupils distribute nominations across peers over time), and the nomination cap (up to 12) combined with a small classroom can limit cumulative advantage. This finding should therefore be interpreted as a network-dynamic pattern captured by the inPop effect under the chosen specification.

Other structural effects and controls. Reciprocity is large and positive (Estimate = 3.604, SE = 0.586, $t = 6.150$), indicating that nominations are much more likely to become mutual than under a baseline random process. The outdegree (density) parameter is negative (Estimate = -0.814 , $t = -1.895$) and mainly captures overall sparsity rather than a substantive mechanism. Outdegree activity is positive (Estimate = 0.127, $t = 2.929$), suggesting heterogeneity in sending nominations.

Reciprocity-related activity (Estimate = -0.460 , $t = -4.218$) indicates that pupils with more reciprocated ties tend to reduce additional outgoing activity.

Model fit and stability: The GoF-based update from Model 1 to Model 2 (adding `reciAct`) substantially improves degree-distribution fit (Table 14; Figures 8–9) while leaving key substantive estimates stable (Table 15). However, the remaining misfit in the triad census (Figure 10; $p = 0.012$) indicates that higher-order local structure is not fully captured, which is a limitation for detailed claims about triad types. Given moderate tie stability and the small constant node set, we prioritize a parsimonious final model and treat residual triadic misfit as an important caveat.

Limitations. The analysis is restricted to 25 pupils observed in all three waves, which reduces statistical power and generalizability. Tie stability is moderate, indicating considerable turnover. Future work could explore additional covariates and alternative structural effects to further refine model fit.